

Separated Phase Registries for Direct Core Twists: A Finite Large-Sieve Gate and a Selector Firewall

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

July 2026

Abstract

TPC-167 gives exact phase- L^2 cancellation for direct additive twists on the literal determinant-two core. We transfer that continuous statement to arbitrary finite separated phase registries. An elementary sampling inequality bounds the registry energy by the coefficient energy times $\delta^{-1} + 4\pi(L-1)$. For a quasi-uniform registry of at least interval-scale size, all but a quantitatively sparse set of phases inherit a fixed- X power saving. A sharp selector example shows why density-one registry control does not identify one distinguished production phase. Thus the finite-registry route genuinely advances, but the original pointwise phase node remains open.

Keywords. large sieve; additive phase; finite registry; selector firewall; Möbius correlation.

Claim boundary. The theorem controls the proportion of bad phases in a separated finite registry. It does not prove that a named production phase is good. The selector counterexample is for the coefficient-blind sampling implication and is not a lower bound for the literal Möbius correlation. There is no physical H3 return, 1/400, prime-pair lower bound, or twin-prime theorem.

1 A separated sampling inequality

Put $e(x) = \exp(2\pi ix)$ and let

$$P(\alpha) = \sum_{n=0}^{L-1} b_n e(n\alpha), \quad E = \sum_{n=0}^{L-1} |b_n|^2.$$

A finite set $\{\alpha_1, \dots, \alpha_M\} \subset \mathbb{T}$ is δ -separated, for a declared $0 < \delta \leq 1$, if

$$\|\alpha_j - \alpha_k\|_{\mathbb{T}} \geq \delta \quad (j \neq k).$$

Theorem 1.1 (Elementary finite phase sieve). *For every $0 < \delta \leq 1$ and every δ -separated registry,*

$$\sum_{j=1}^M |P(\alpha_j)|^2 \leq \{\delta^{-1} + 4\pi(L-1)\}E. \tag{1}$$

Proof. Let I_j be the circular interval of length δ centered at α_j . These intervals have disjoint interiors. With $g = |P|^2$, for $x \in I_j$,

$$g(\alpha_j) \leq g(x) + \int_{I_j} |g'(y)| dy.$$

Integrate in x over I_j , sum in j , and divide by δ :

$$\sum_j g(\alpha_j) \leq \delta^{-1} \int_{\mathbb{T}} |P|^2 + \int_{\mathbb{T}} |(|P|^2)'|.$$

Parseval gives $\int |P|^2 = E$, while Cauchy–Schwarz gives

$$\int_{\mathbb{T}} |(|P|^2)'| \leq 2\|P'\|_2\|P\|_2 \leq 4\pi(L-1)E.$$

This proves (1). □

The constant is not optimized; the point is an explicit standard-library-auditable inequality. Classical large-sieve inequalities can sharpen constants without changing the route semantics [1].

2 The determinant-two registry theorem

Retain

$$\begin{aligned} q &= as, \quad t(z) = ad + qz, \quad c_z = \mu(d + sz)\mu(u + az), \\ I_N &= \{z : N < t(z) \leq 2N\}, \quad L_N = |I_N|, \quad E_N = \sum_{I_N} |c_z|^2, \end{aligned}$$

and

$$F_N(\alpha) = \frac{q}{N} \sum_{z \in I_N} c_z e(-\alpha z).$$

Translation of the consecutive exponent interval changes only a unit-modulus factor, so [theorem 1.1](#) applies.

Corollary 2.1 (Bad-phase count). *For every $\lambda > 0$, the number B_λ of registry phases with $|F_N(\alpha_j)| > \lambda$ satisfies*

$$B_\lambda \leq \frac{q^2 E_N}{N^2 \lambda^2} \{\delta^{-1} + 4\pi(L_N - 1)\}. \quad (2)$$

Proof. Each bad phase contributes more than λ^2 to the sampled normalized energy. Apply [theorem 1.1](#). □

Corollary 2.2 (Quasi-uniform registry density). *Suppose*

$$\delta \geq \frac{c}{M}, \quad M \geq \theta L_N$$

for fixed $c, \theta > 0$. Then

$$\frac{B_\lambda}{M} \leq \left(c^{-1} + \frac{4\pi}{\theta}\right) \frac{q/N + q^2/N^2}{\lambda^2}. \quad (3)$$

If $q/N \rightarrow 0$, the choice $\lambda = (q/N)^{1/4}$ makes the bad proportion $O_{c,\theta}((q/N)^{1/2})$.

Proof. Divide (2) by M , use $\delta^{-1}/M \leq c^{-1}$, $L_N/M \leq \theta^{-1}$, and

$$E_N \leq L_N \leq N/q + 1.$$

□

In the small-polylogarithmic envelope with $N \geq \sqrt{X}$, this threshold is at most

$$X^{-1/8}(\log X)^{\eta_0/4}.$$

The theorem therefore has the typed program status

PROVED_L1_ACTUAL_CORE_PHASE_METRIC_FINITE_REGISTRY.

axis	value
analytic_norm	L2_PHASE_REGISTRY
program_positive_L2	false
fixed_atom	false

The power is a density statement on a certified registry, not a uniform pointwise estimate.

3 The selector firewall

Proposition 3.1 (Density does not select). *There are quasi-uniform registries and coefficient sequences for which the bad proportion tends to zero while a distinguished registry phase has normalized value one at every length.*

Proof. Take $b_n = 1$ for $0 \leq n < L$ and the complete grid $\alpha_r = r/L$. Finite Fourier orthogonality gives

$$\frac{1}{L} \left| \sum_{n=0}^{L-1} e(n\alpha_r) \right| = \mathbf{1}_{\{r=0\}}.$$

Thus the bad proportion at any fixed threshold in $(0, 1)$ is $1/L$, but the named phase $\alpha_0 = 0$ remains maximally bad. $\square$

This is a scoped logical stop:

density-one registry control $\not\Rightarrow$ control of a named selector.

The coefficients in [proposition 3.1](#) are not asserted to be the literal Möbius coefficients. The proposition stops only a coefficient-blind selection inference. It does not say that any minor-arc Möbius twist is large, just as the projection obstruction in TPC-158 stops only its declared approximation route [\[2\]](#).

4 Route decision and next object

TPC-167 supplied continuous phase L^2 [\[3\]](#); TPC-168 now supplies finite separated-registry L^2 . This is useful if the production observable averages over a certified phase population. A distinguished phase requires one of the following additional objects:

- (i) a source-locked selector crosswalk proving that the chosen phase avoids the sparse bad subset;
- (ii) a random or equidistributed selector theorem with the exact quantifiers needed by the application;
- (iii) a pointwise additive-twist theorem.

None is currently present. The original parent-ready pointwise node therefore remains open rather than stopped.

5 Reproducible audit

The accompanying script source-locks TPC-167, checks (1) and (2) on a nonuniform separated fixture, and realizes [proposition 3.1](#) on a complete grid. Executable bad-snapshot mutations reject missing separation, promotion from density one to all phases, and interpretation of the selector fixture as a Möbius lower bound. Hashes have integrity semantics only.

6 Conclusion

Phase averaging now survives discretization: a sufficiently dense separated registry has only a quantitatively sparse set of large direct twists. The result is a new actual-core phase-metric gate whose analytic norm is L^2 in the registry variable; it is not the program's positive- $L2$ parity gate. The selector firewall identifies the exact remaining gap, rather than hiding it in an average. TPC-169 turns to the other open door and asks whether one common phase-exceptional set can control every atomic prefix endpoint.

References

- [1] Hugh L. Montgomery and Robert C. Vaughan. *Multiplicative Number Theory I: Classical Theory*. Cambridge University Press, 2007.
- [2] Liang Wang. An Exact Major-Arc Gate for Additive Phases: Periodic Projection and a Minor-Arc Route Obstruction, 2026. TPC-158 manuscript.
- [3] Liang Wang. A Parseval Corridor for Direct Additive Twists on the Determinant-Two Möbius Core, 2026. TPC-167 manuscript.